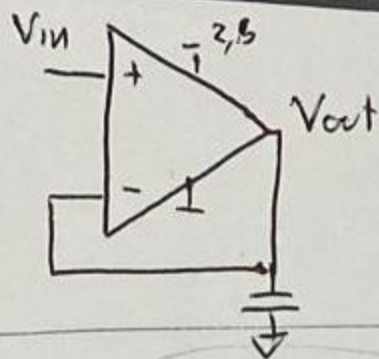
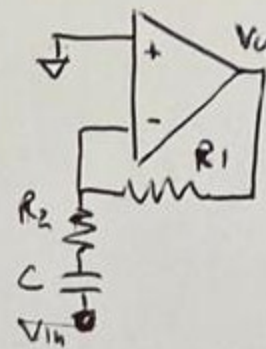


① Giderle



→ INSEOUTONİ
→ Giderle = 1

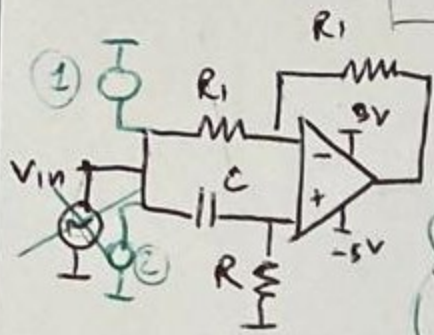
10/05/21



$$\frac{V_{out}}{R_1} = \frac{-V_{in}}{R_2 + \frac{1}{sC_1}}$$

$$\frac{V_{out}}{V_{in}} = \frac{-R_1 s C_1}{(R_2 s C_1 + 1)}$$

13/06/22



SDE

①: $\frac{V_1}{R_1} = -\frac{V_0}{R_1} \Rightarrow V_1 = -V_0$

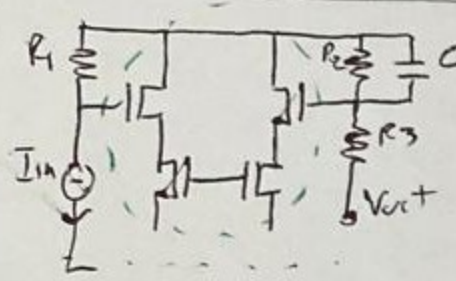
②: $\frac{V_0}{2} = V_1 \cdot \frac{R}{R + sC}$

+ = Giderle = $\frac{2R}{R + \frac{1}{sC}} - 1 = \frac{R s C - 1}{R s C + 1}$

spektr x 10

Giderle = 10

14/06/21



Giderle = $\frac{V_{out}}{I_{in}}$

$I_{in} \cdot R_1 = I_{out} (R_2 + \frac{1}{sC_1})$

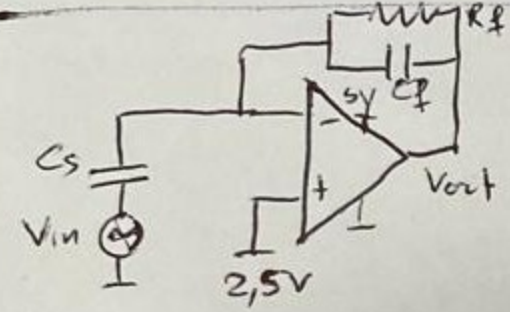
$\frac{I_{out}}{I_{in}} = \frac{R_1 (1 + sC_1 R_2)}{R_2}$

$\frac{V_{out} + I_{in} \cdot R_1}{R_2} = -\frac{R_1 (1 + sC_1 R_2)}{R_2} I_{in}$

$V_{out} = -\frac{R_3 R_1 (1 + sC_1 R_2)}{R_2} I_{in} - I_{in} R_1$

$\frac{V_{out}}{I_{in}} = -R_1 (1 + \frac{R_3 (1 + sC_1 R_2)}{R_2})$

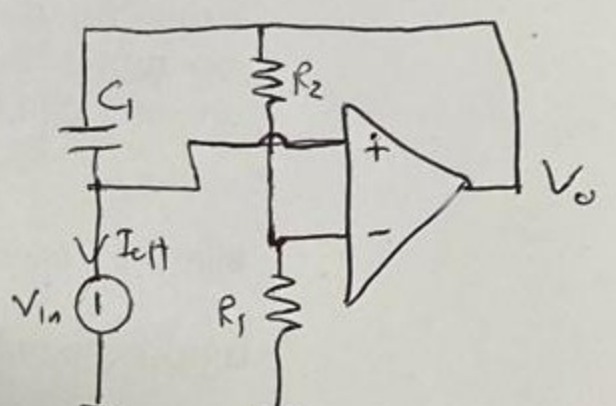
15/07/19



$\frac{V_{out}}{R_f + \frac{1}{sC_f}} = -\frac{V_{in} s C_s}{R_f + \frac{1}{sC_f}}$

$\frac{V_{out}}{V_{in}} = \frac{-s C_s R_f}{s C_f R_f + 1}$

17/01/23



Giderle = $\frac{I_{out}(s)}{V_{in}(s)}$

$\frac{V_{in}}{R_1} \cdot R_1 + R_2 = V_0$

$(\frac{R_1 + R_2}{R_1} - 1) V_{in} s C = I_{out}$

$\frac{I_{out}}{V_{in}} = \frac{R_2 s C_1}{R_1}$

17/02/20

$\frac{I_{out}}{I_{in}} \rightarrow V_0 = V_i$

$\frac{V_i}{R_1} = I_{in} = \frac{V_0}{R_1}$

$\frac{V_0}{R_2} = I_{out} \rightarrow V_0 = R_2 I_{out}$

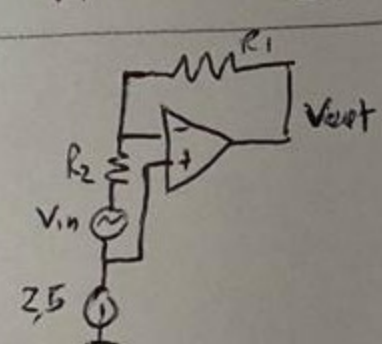
$I_{in} = \frac{V_0}{R_1} = \frac{R_2}{R_1} I_{out}$

$\frac{I_{out}}{I_{in}} = \frac{R_1}{R_2}$

17/06/19

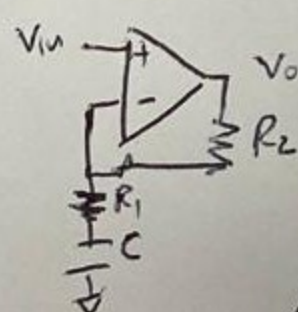
Giderle → (INSEOUTONİ)
Giderle = 1

22/02/21



$\frac{V_{out}}{R_1} = -\frac{V_{in}}{R_2} \rightarrow \frac{V_{out}}{V_{in}} = -\frac{R_1}{R_2} = -\frac{1}{2}$

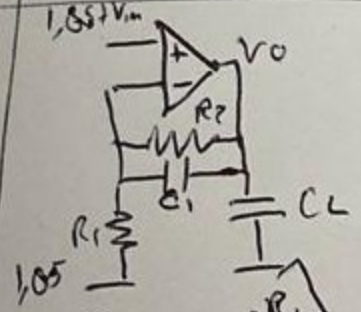
22/09/21



$V_{in} = V_0 \cdot \frac{R_1 + sC_1}{R_2 + R_1 + \frac{1}{sC_1}}$

$\frac{V_0}{V_{in}} = \frac{1 + (R_1 + R_2) s C_1}{(R_1 + sC_1 + 1)}$

23/01/2020



$\frac{V_{in}}{R_1} \left[\frac{R_2 + \frac{1}{sC_1}}{R_1 + sC_1 + 1} \right] = V_0$

$\frac{V_0}{V_{in}} = \frac{R_2 + R_1 + sC_1 R_2 R_1}{R_1 (1 + sC_1 R_2)}$

27/01/21

$V_{in} = V_{out} \cdot \frac{R_2 + sC_1}{R_1 + \frac{1}{sC_1} + R_1}$

